Supplementary Information

Thermal Shielding, Melt Kinetics and Surge Control of a Steam-Assisted Pipeline Cleaning Crawler: A Monte Carlo Study and Its Deployment Limits

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S1. Thermal sweep (first suite, 3,000 draws per case; 800 per thickness)

Aerogel (mm)	Atmospheric, 45 min	Atmospheric, 90 min	Evacuated, 45 min	Evacuated, 90 min
5	0.1% (0.0–0.7%)	0%	51.9% (48.4–55.3%)	0%
10	19.1% (16.5–22.0%)	0%	92.4% (90.3–94.0%)	12%
15	62.4% (59.0–65.7%)	0%	97.1% (95.7–98.1%)	35%
20	86.5% (84.0–88.7%)	1%	98.6% (97.6–99.2%)	50%
30	96.2% (94.7–97.4%)	16%	98.9% (97.9–99.4%)	69%
40	99.1% (98.2–99.6%)	40%	99.6% (98.9–99.9%)	80%
50	99.2% (98.4–99.7%)	60%	99.6% (98.9–99.9%)	80%
60	100.0% (99.5–100.0%)	72%	98.9% (97.9–99.4%)	79%

Table S1. P(pack ≤ 55 °C) against aerogel thickness, 45 min (Wilson 95% interval) and 90 min.

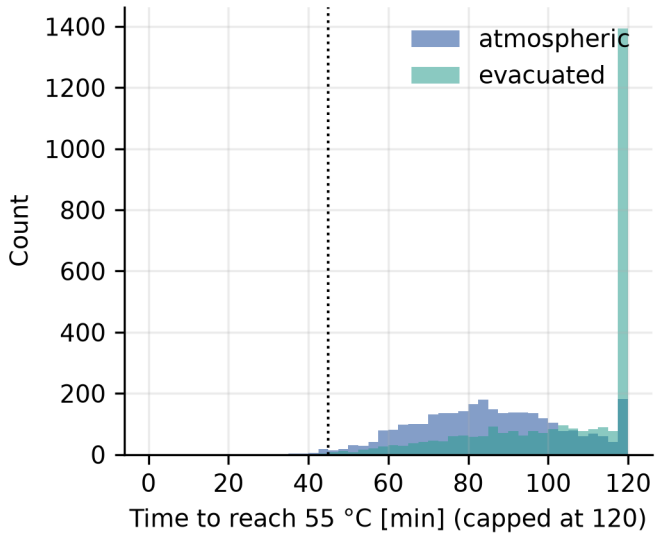


Fig. S1. Time to reach 55 °C for atmospheric and evacuated aerogel, 40 mm (capped at 120 min).

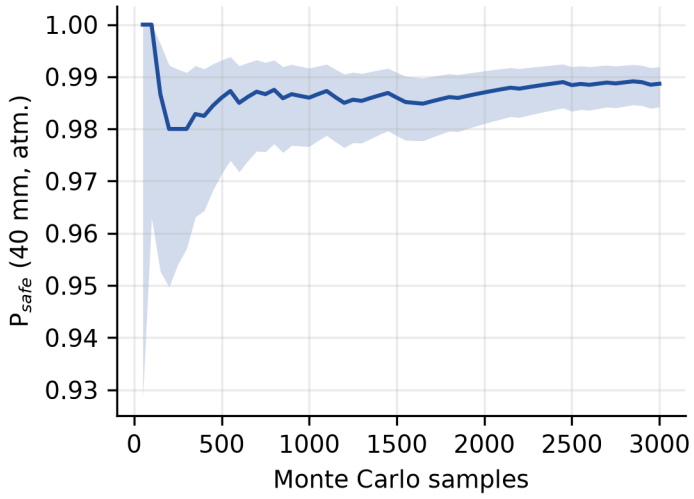


Fig. S2. Convergence of the Monte Carlo estimate of P(pack ≤ 55 °C) with the number of draws (40 mm, atmospheric); band: Wilson 95% interval.

S2. Pipe length and logistics (all cases)

Mission	Melt removal	L (m)	P(one)	Dep. median	Dep. p95	Active (h)	Total (h)	Range (m)
45 min atmospheric	ablation	0.3	100%	1	1	0.00	11.5 (8.3–14.7)	133.6
45 min atmospheric	ablation	10	100%	1	1	0.06	11.6 (8.4–14.8)	133.6
45 min atmospheric	ablation	100	65%	1	3	0.56	12.9 (9.1–18.4)	133.6
45 min atmospheric	ablation	300	0%	3	8	1.68	17.1 (12.0–31.5)	133.6
45 min atmospheric	ablation	600	0%	5	16	3.36	22.6 (14.9–51.9)	133.6
45 min atmospheric	ablation	1,000	0%	8	26	5.61	30.6 (18.2–79.6)	133.6
45 min atmospheric	retained 0.5 mm	0.3	100%	1	1	0.02	11.5 (8.3–14.7)	12.5
45 min atmospheric	retained 0.5 mm	10	62%	1	3	0.60	13.1 (9.1–18.5)	12.5
45 min atmospheric	retained 0.5 mm	100	0%	9	27	6.00	32.1 (16.8–81.2)	12.5
45 min atmospheric	retained 0.5 mm	300	0%	25	79	18.01	75.3 (30.6–224.9)	12.5
45 min atmospheric	retained 0.5 mm	600	0%	49	158	36.02	140.0 (50.0–438.8)	12.5
45 min atmospheric	retained 0.5 mm	1,000	0%	81	263	60.03	227.1 (76.3–723.3)	12.5
45 min atmospheric	no drainage	0.3	97%	1	1	0.10	11.7 (8.5–15.1)	2.3
45 min atmospheric	no drainage	10	9%	5	28	3.30	22.7 (11.6–82.2)	2.3
45 min atmospheric	no drainage	100	0%	45	274	32.99	130.5 (29.2–725.2)	2.3
45 min atmospheric	no drainage	300	0%	133	822	98.96	369.0 (66.2–2,154.9)	2.3
45 min atmospheric	no drainage	600	0%	265	1,643	197.92	727.6 (121.5–4,301.2)	2.3
45 min atmospheric	no drainage	1,000	0%	442	2,738	329.86	1,206.4 (195.9–7,162.3)	2.3
90 min evacuated	ablation	0.3	100%	1	1	0.00	11.5 (8.3–14.7)	255.2
90 min evacuated	ablation	10	100%	1	1	0.06	11.6 (8.4–14.8)	255.2
90 min evacuated	ablation	100	89%	1	2	0.56	12.4 (9.0–16.2)	255.2
90 min evacuated	ablation	300	41%	2	5	1.68	14.9 (10.1–24.5)	255.2
90 min evacuated	ablation	600	0%	3	9	3.37	18.8 (13.0–38.2)	255.2
90 min evacuated	ablation	1,000	0%	4	14	5.61	23.9 (15.0–57.0)	255.2
90 min evacuated	retained 0.5 mm	0.3	100%	1	1	0.02	11.5 (8.3–14.7)	23.8
90 min evacuated	retained 0.5 mm	10	89%	1	2	0.60	12.4 (9.0–16.3)	23.8
90 min evacuated	retained 0.5 mm	100	2%	5	14	6.02	24.9 (14.4–57.8)	23.8
90 min evacuated	retained 0.5 mm	300	0%	13	42	18.05	53.4 (23.9–151.6)	23.8
90 min evacuated	retained 0.5 mm	600	0%	26	84	36.10	95.9 (37.1–292.9)	23.8
90 min evacuated	retained 0.5 mm	1,000	0%	43	140	60.17	152.5 (54.9–480.9)	23.8
90 min evacuated	no drainage	0.3	99%	1	1	0.10	11.7 (8.4–14.9)	4.3

Mission	Melt removal	L (m)	P(one)	Dep. median	Dep. p95	Active (h)	Total (h)	Range (m)
90 min evacuated	no drainage	10	23%	3	15	3.29	18.6 (10.6–58.7)	4.3
90 min evacuated	no drainage	100	0%	24	146	32.89	88.6 (23.2–498.1)	4.3
90 min evacuated	no drainage	300	0%	70	438	98.66	245.1 (47.9–1,473.8)	4.3
90 min evacuated	no drainage	600	0%	138	875	197.31	479.7 (84.6–2,936.8)	4.3
90 min evacuated	no drainage	1,000	0%	230	1,458	328.86	792.3 (134.8–4,891.9)	4.3

Table S2. Deployments, active time and total intervention time by mission, melt-removal case and pipe length (median; total with 5th–95th percentile).

S3. Pipe size and temperature (detail)

NPS (in)	Bore (mm)	Hull OD (mm)	Pack radius (mm)	Aerogel (mm)	P atm.	P evac.	h scale	Steam scale
4	102.3	–	25	does not fit	–	–	1.31	0.34
6	154.1	65	25	does not fit	–	–	1.19	0.51
8	202.7	114	25	24	0.0% (0.0–0.5%)	5.1% (3.8–6.9%)	1.11	0.67
10	254.5	166	35	40	42.2% (38.9–45.7%)	50.0% (46.5–53.5%)	1.05	0.83
12	304.8	216	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	1.00	1.00
14	336.5	248	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	0.98	1.10
16	387.3	298	60	40	99.2% (98.4–99.7%)	99.4% (98.5–99.7%)	0.94	1.27

Table S3. Fit and thermal margin by nominal pipe size (800 draws; Wilson interval). h scale: condensation coefficient relative to 12 in; steam scale: melting steam per metre relative to 12 in.

Temperature (°C)	P atm. (95% interval)	P evac. (95% interval)	Median pack T at 45 min (°C)	Melt time (s)	Range (m)
0	100.0% (99.5–100.0%)	100.0% (99.5–100.0%)	14.3	13.1	111
10	100.0% (99.5–100.0%)	100.0% (99.5–100.0%)	24.2	12.3	119
25	98.8% (97.7–99.3%)	99.9% (99.3–100.0%)	39.1	10.9	133
35	84.5% (81.8–86.8%)	89.8% (87.5–91.7%)	49.0	10.1	139
45	21.9% (19.1–24.9%)	35.9% (32.6–39.3%)	58.9	9.3	102
55	0.4% (0.1–1.1%)	0.1% (0.0–0.7%)	68.7	8.4	0

Table S4. Pipe / fluid temperature (12 in design, 40 mm aerogel; 800 draws): probability, median pack temperature at 45 min (atmospheric), median melt time and range.

S4. Bypass: peak speed by pocket volume

V_g (m³)	0	5	10	20	40	80	160	320
0.3	97%	97%	96%	95%	92%	88%	82%	72%
1.0	98%	98%	98%	98%	97%	95%	92%	87%
3.0	99%	99%	99%	98%	98%	98%	96%	94%
10.0	99%	99%	99%	99%	99%	99%	98%	98%

Table S5. P(peak speed > 1 m/s) against bypass area (cm²; columns) for four gas-pocket volumes (rows, m³); other inputs sampled.

V_g (m³)	20 cm²	80 cm²	320 cm²
0.3	76	19	5
1.0	254	64	16
3.0	762	190	48
10.0	2,538	635	159

Table S6. Median time for a fully open bypass to halve the break-through differential (ms), tool held at rest, against pocket volume and bypass area.

Q_s (m³/s at upstream conditions)	Bypass area (cm²)
0.01	1.6

Q_s (m³/s at upstream conditions)	Bypass area (cm²)
0.03	4.7
0.10	15.6
0.30	46.9
1.00	156.3

Table S7. Bypass flow area that holds a 0.5 bar differential at 18 bar upstream (assumed methane-like gas, C_d = 0.7) for a supply flow Q_s.

S5. Automated checks

tests/test_all.py runs 27 checks: the first suite is unchanged (sha256) and reproduces its published results; the maintenance-lattice formula; the extended thermal, melt and surge functions equal the first suite when the added mechanisms are off (pack temperature histories, speed, range, steam per metre and melt time, peak speeds for the passive and controller-only crawlers, and the zero-area orifice); monotonic range with retained film; the no-drainage limit; the series-resistance flux; the $D^{-1/4}$ scaling of the condensation coefficient; zero and choked orifice flow; the gas fraction bounds; monotonic bypass benefit; venting-time scaling with volume; proportionality of the holding area to the supply flow; the 12 in hull; the fit in 4, 6 and 8 in pipes; the wall-heating energy balance; the standard-wall pipe mass; the baseline probability against the first suite's interval; monotonic temperature dependence; mesh convergence; and the 270 m and 540 m range caps implied by the speed set-point.

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